MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 Admission Onwards)

Minor Course- MG1DSCBCH100- Biochemistry-The Science of Life

Duration: 1.5 Hours Maximum Marks: 50

Students should attempt at least one question from each outcome to enhance their overall outcome attainability

Part A
Multiple Choice Questions
(Answer all questions; each question carries 1 mark)

1. Identify the component present in nucleic acids. [U] [4]
   (a) Glucose (b) Glycine (c) Adenine (d) Stearic acid
2. The bond angle of water is [K] [1]
   (a) 100.5^0 (b) 104.5^0 (c) 105.4^0 (d) 101.5^0
3. Recall an example for a monosaccharide [K] [2]
   (a) Lactose (b) Mannose (c) Maltose (d) Sucrose
4. The single letter code of tyrosine [U] [3]
   (a) G (b) Y (c) T (d) R
5. Cholesterol belongs to which of these lipids [E] [4]
   (a) Phospholipid (b) Glycolipid (c) Steroid (d) None of these

(1x 5= 5)

Part B
Fill in the Banks
(Answer all questions; each question carries 2 marks)

6. In addition to glycerol and fatty acids ---------is also present in phosphatidic acid. [K] [4]
7. An aminoacid which carries a positive as well as negative charge in its structure is said to exist as ……. . [U] [3]
8. The theory which states that living organisms obey the laws of Physical and Chemical Sciences is called ---------- [U] [1]
9. The α and β forms of glucopyranoses are called as …….. [K] [2]
10. The concentration of ----------ions is greater than … ions in an acidic solution. [U] [1]

(2 x 5=10)

Part C
Short Answer Questions
(Answer any 5 questions; each question carries 3 marks)

11. Assess the action of bicarbonate buffer [I] [1]
12. Classify carbohydrates based on their chemical structure [K] [2]
13. Explain in which all areas Biochemistry find applications. [U] [1]
14. Evaluate the significance of phosphodiester linkage in the stability of nucleic acids. [E] [4]
15. Illustrate the structure of cholesterol and give its functions. [K] [4]
16. What are mesosomes? Give the functions. [K] [1]
17. Differentiate alpha helix and beta pleated sheets. [E] [3]

(3 x 5=15)

Part D
Short Essay Type Question
(Answer any four questions; each question carries 5 marks)

18. Describe the structural organization of proteins. [K] [3]
19. Examine the structure of B -DNA. [E] [4]
20. Give a detailed report of aldose sugars with structures of any three. [U] [2]
21. Explain the physical and chemical properties of water. [U] [1]
22. Discuss the scope of Biochemistry in the light of advancements in technology. [U] [1]
23. Illustrate the Haworth structure of disaccharides. [U] [2]

(5 x 4 = 20)